

Project Q&A Hub

This directory contains technical interview questions and answers directly mapped to the source code implementations of the three active projects in the workspace.

Directory Index

[project_1_qa.md](#)

- **Topic:** Production-grade Retrieval-Augmented Generation (RAG) Q&A System.
- **Key Concepts Covered:**
 - End-to-end data flow (Loader, splitter, caching, FAISS, LCEL chains).
 - Strict anti-hallucination prompt formatting and post-inference refusal checks.
 - On-disk embedding cache (`CacheBackedEmbeddings` + `LocalFileStore`).
 - Question condensation using historical conversational memory.
 - **Architecture Diagram:** RAG ingestion and inference loop.

[project_2_qa.md](#)

- **Topic:** SentinelGuard – Time-Series Anomaly Detection System.
- **Key Concepts Covered:**
 - Overall system pipeline (S&P 500 log returns, training split).
 - Clean normal sequence training filters based on variance thresholds.
 - LSTM Autoencoder layers configuration (`RepeatVector` , `TimeDistributed`).
 - Anomaly point post-processing (Noise Floor, Dynamic Percentile, Temporal Smoothing).
 - **Architecture Diagram:** Data ingestion, autoencoder training, and inference pipelines.

[project_3_qa.md](#)

- **Topic:** Retail Sales Forecasting Pipeline.
- **Key Concepts Covered:**
 - End-to-end modular pipeline (Data aggregation, features, evaluation, UI).
 - Feature engineering strategy (Time, memory lags, rolling stats, momentum).
 - Chronological train-test splitting vs. standard random K-Fold CV.
 - Random Forest vs. XGBoost model comparison and R^2 variance interpretation.

- **Architecture Diagram:** Data loader, feature merging, training, and evaluation pipeline.